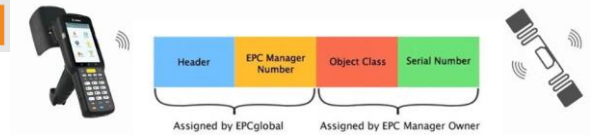
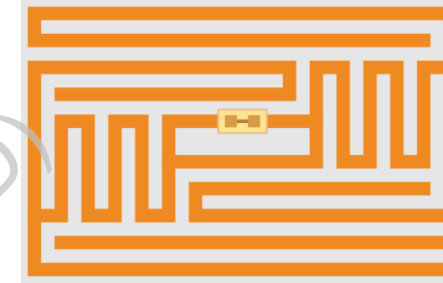


Wireless IoT Network Protocols



IoT



Networking

Prof. Mohd. Iqbal Bhat
Government Degree College Beerwah

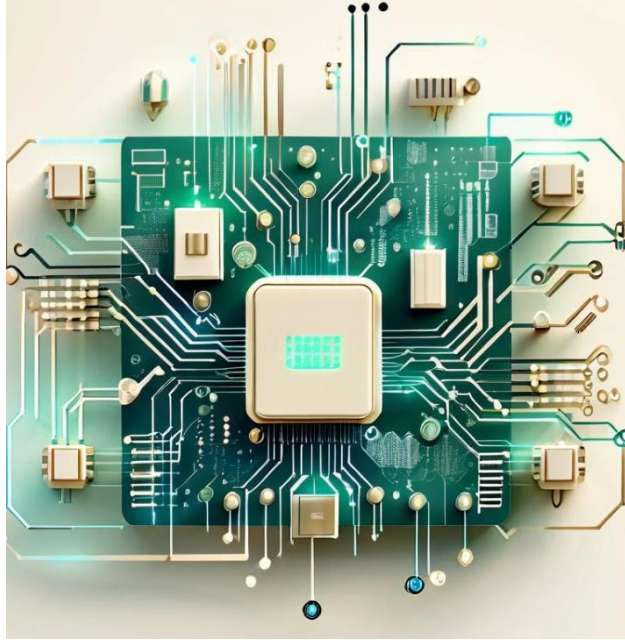
CoAP

Functionality-based IoT Protocol Organization

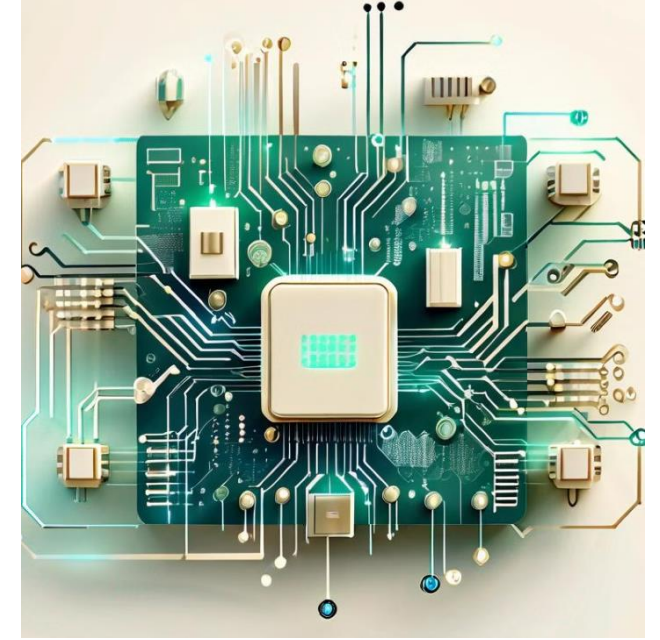
- ✓ **Connectivity** (6LowPAN, RPL)
- ✓ **Identification** (EPC, uCode, IPv6, URIs)
- ✓ **Communication / Transport** (WiFi, Bluetooth, LPWAN)
- ✓ **Discovery** (Physical Web, mDNS, DNS-SD)
- ✓ **Data Protocols** (MQTT, CoAP, AMQP, Websocket, Node)
- ✓ **Device Management** (TR-069, OMA-DM)
- ✓ **Semantic** (JSON-LD, Web Thing Model)
- ✓ **Multi-layer Frameworks** (Alljoyn, IoTivity, Weave, Homekit)

Source: [Internet of Things Protocols \(Online\)](#)

Data Protocols



CoAP



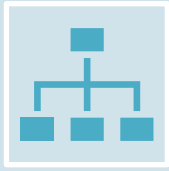
CoAP (Constrained Application Protocol) is a lightweight, UDP-based protocol designed for resource-constrained devices and low-power networks. It enables efficient communication between IoT sensors, actuators, and servers, making it well-suited for applications like smart homes, industrial automation, and environmental monitoring.

Introduction

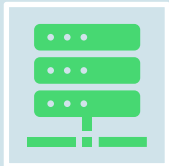
- ✓ CoAP – **Constrained Application Protocol**.
- ✓ **IoT Web transfer protocol** for use with constrained nodes and networks.
- ✓ **Designed for Machine to Machine (M2M)** applications such as smart energy and building automation.
- ✓ Based on **Request-Response model** between end-points
- ✓ Client-Server interaction is **asynchronous over a datagram oriented transport protocol** such as UDP
- ✓ Uses fewer Resources than HTTP

- ✓ The Constrained Application Protocol (CoAP) is a session layer protocol designed by IETF Constrained RESTful Environment (CoRE) working group to provide lightweight RESTful (HTTP) interface.
- ✓ Representational State Transfer (REST) is the standard interface between HTTP client and servers.
- ✓ Lightweight applications such as those in IoT, could result in significant overhead and power consumption by REST.
- ✓ CoAP is designed to enable low-power sensors to use RESTful services while meeting their power constraints.

Key Characteristics of CoAP



RESTful Design: CoAP adheres to the REST (Representational State Transfer) architectural style, employing resources identified by Uniform Resource Identifiers (URIs) and manipulated through methods like GET, POST, PUT, and DELETE. This facilitates interoperability and simplifies development.



UDP Binding: Unlike HTTP, which primarily relies on TCP, CoAP utilizes the User Datagram Protocol (UDP). UDP's connectionless nature reduces overhead, making it suitable for constrained devices and low-power networks.



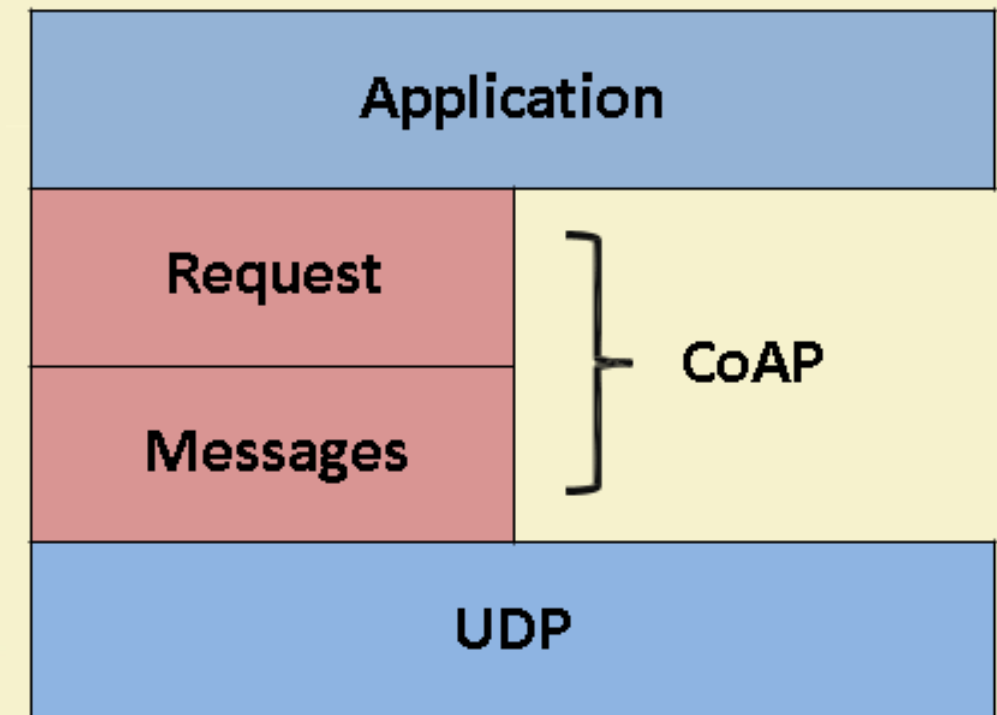
Lightweight Messaging: CoAP messages are compact, with minimal header information, reducing bandwidth consumption and processing requirements.



Asynchronous Communication: CoAP supports asynchronous message exchanges, enabling efficient communication for time-sensitive applications and event-driven architectures.

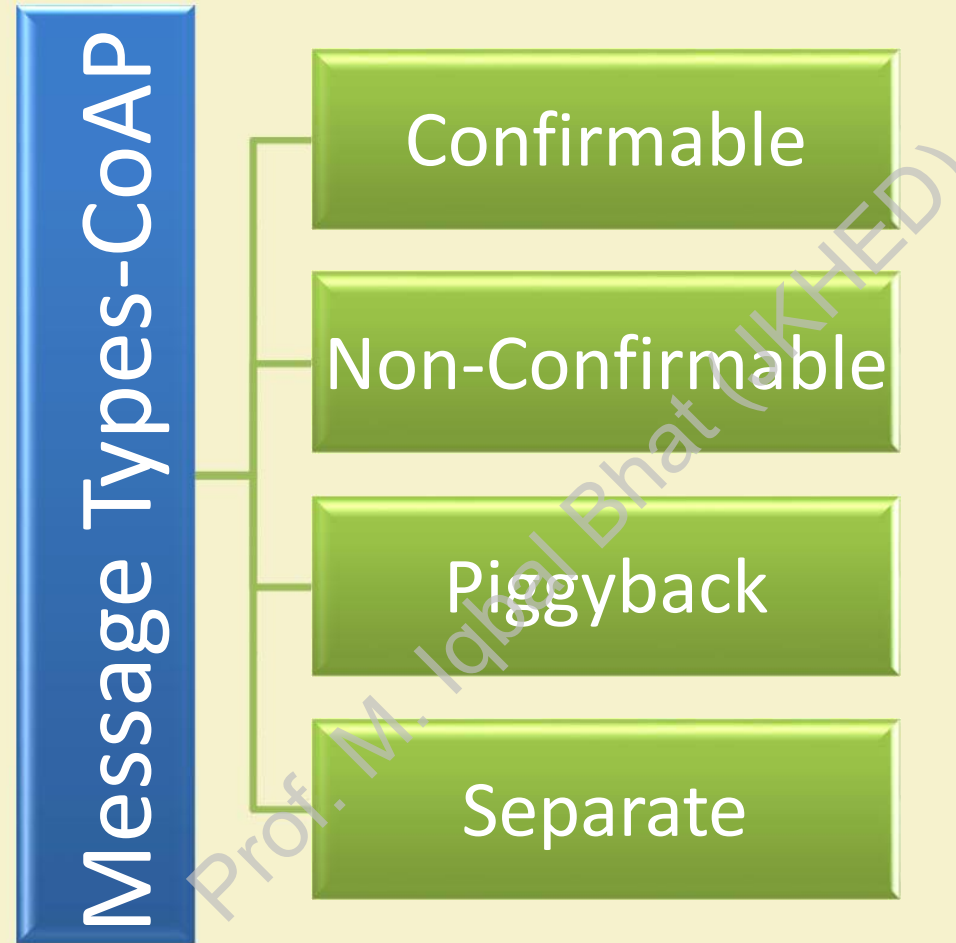
Architecture of CoAP

- ✓ CoAP architecture is divided into two main sub-layers:
 - Messaging
 - Request/response.
- ✓ The messaging sub-layer is responsible for reliability and duplication of messages, while the request/response sub-layer is responsible for communication.
- ✓ CoAP has four messaging modes:
 - Confirmable
 - Non-confirmable
 - Piggyback
 - Separate



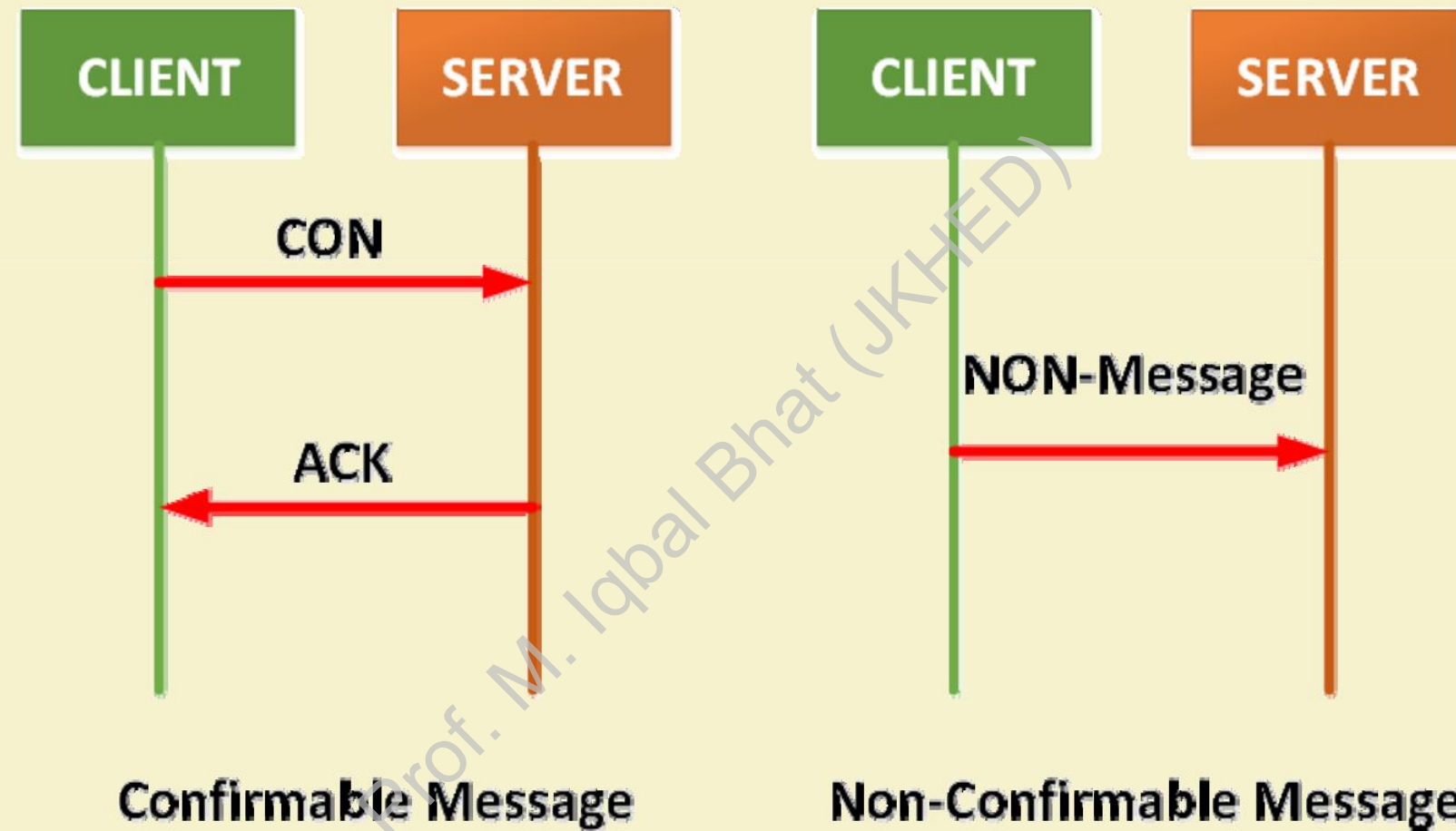
Source: V. Karagiannis, P. Chatzimisios, F. Vazquez-Gallego, and J. Alonso-Zarate, "A survey on application layer protocols for the internet of things," Transaction on IoT and Cloud Computing, vol. 3, no. 1, pp. 11-17, 2015

CoAP Message Types



Source: Z. Shelby , K. Hartke, C. Bormann, **“The Constrained Application Protocol (CoAP)”**, Internet Engineering Task Force (IETF), Standards Track, 2014

CoAP Request-Response Model

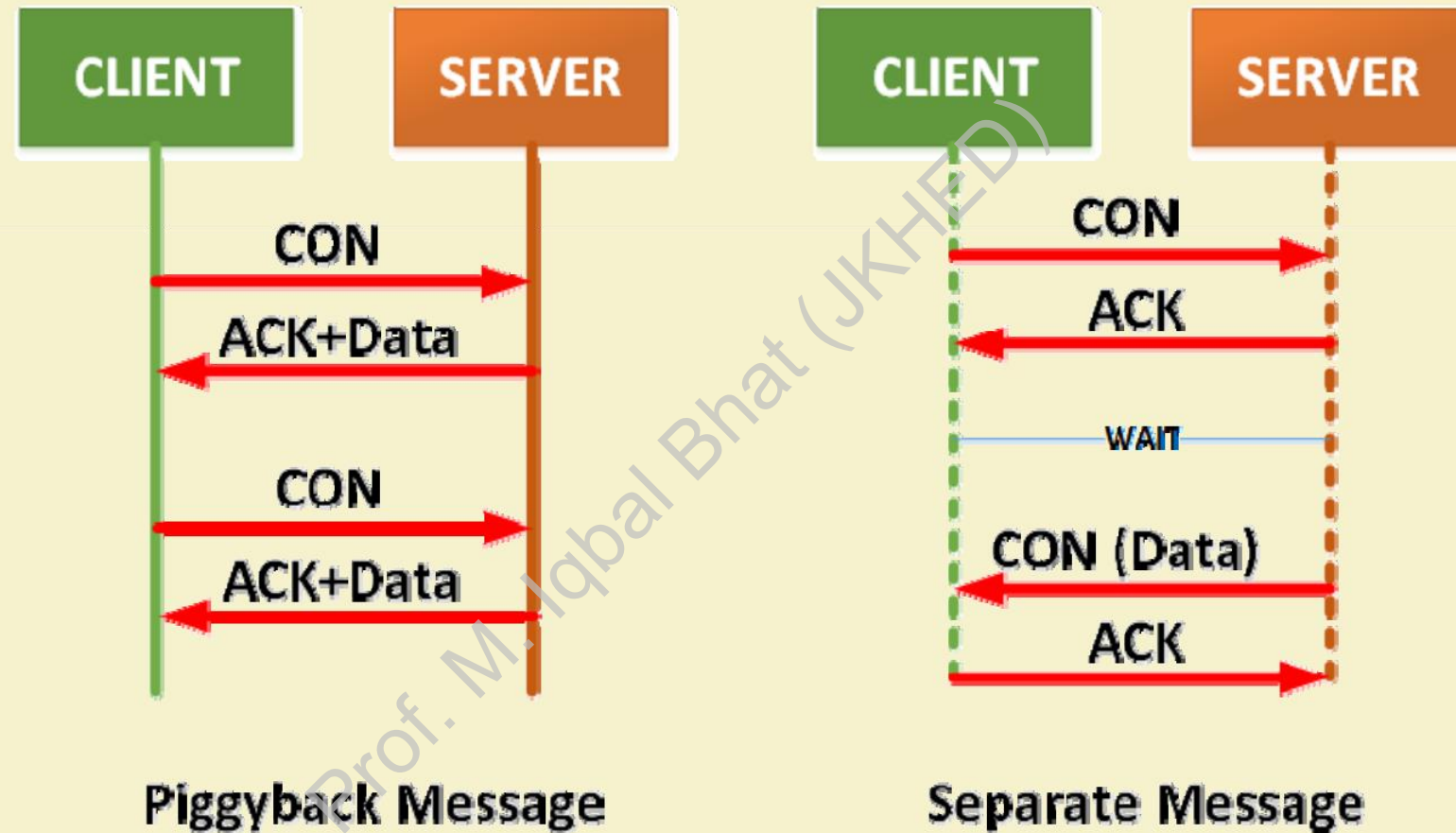


Source: V. Karagiannis, P. Chatzimisios, F. Vazquez-Gallego, and J. Alonso-Zarate, "A survey on application layer protocols for the internet of things," Transaction on IoT and Cloud Computing, vol. 3, no. 1, pp. 11-17, 2015

- ✓ Confirmable and non-confirmable modes represent the reliable and unreliable transmissions, respectively, while the other modes are used for request/response.
- ✓ Piggyback is used for client/server direct communication where the server sends its response directly after receiving the message, i.e., within the acknowledgment message.
- ✓ On the other hand, the separate mode is used when the server response comes in a message separate from the acknowledgment, and may take some time to be sent by the server.
- ✓ Similar to HTTP, CoAP utilizes GET, PUT, PUSH, DELETE messages requests to retrieve, create, update, and delete, respectively

Source: V. Karagiannis, P. Chatzimisios, F. Vazquez-Gallego, and J. Alonso-Zarate, "A survey on application layer protocols for the internet of things," Transaction on IoT and Cloud Computing, vol. 3, no. 1, pp. 11-17, 2015

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Features

- ✓ Reduced overheads and parsing complexity.
- ✓ URL and content-type support.
- ✓ Support for the discovery of resources provided by known CoAP services.
- ✓ Simple subscription for a resource, and resulting push notifications.
- ✓ Simple caching based on maximum message age.

